Simple Expressions for Success Run Distributions in Bernoulli Trials

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Abstract

New simple formulae for some probability distributions of success runs in Bernoulli trials are found by using the classical definition of run. These expressions contain only one summation of ordinary binomial coefficients and thus allow a faster and efficient computation.

Keywords: Bernoulli trials, number of success runs, longest success run, discrete distributions of order k.

1 Introduction

Most recent studies on success runs in Bernoulli trials follow the framework contained in the fundamental book of Feller (1968) and in particular his definition of run as a recurrent pattern. According to this definition two consecutive success runs may not be separated by any failure. As an example, the sequence SSSSSS (where the symbol S denotes a success) can be interpreted as containing 3 success runs of length 2 or 2 success runs of length 3. In practice, if we search for runs of length k, the counting of consecutive successes must be restarted when the desired value k is reached (see Feller, 1968, pag. 305).

It follows from this definition that the location of success runs in a sequence of n Bernoulli trials depends on the reference length k. Although this can seem quite unnatural, some mathematical derivations are greatly simplified particularly when dealing with asymptotical expressions. Moreover, in some cases, such as the probability distribution for the longest run, the final relation does not depend on the definition employed during the proof.

In the present work the study of success runs in Bernoulli trials is carried out by using the classical definition which asserts that two consecutive runs must be separated by one or more failures. Following this approach in section 2 basic expressions for $P(M_n^{(k)} = x)$ and $P(L_n \leq k-1)$ are derived, where $M_n^{(k)}$ is the number of success runs with length k or more and L_n is the length of the longest success run in n Bernoulli trials. Unlike corresponding formulae obtained by Philippou and Makri (1986) and Hirano (1986), only ordinary (first order) binomial coefficients are employed and summations over an index set determined by the solutions of a diophantine equation are not involved.

In particular, the expression for the distribution of L_n reported in Burr and Cane (1961) and Godbole (1990) is again obtained by following a new procedure which allows to find simpler formulae containing a single summation (section 3). Such an approach can be extended to the derivation of probability distributions of similar random variables, such as the kth order negative binomial and the kth order geometric ones (introduced by Philippou, Georghiou and Philippou, 1983). Their interest from a computational point of view is evident.

2 Basic expressions for the distribution of $M_n^{(k)}$ and L_n

Referring to the classical definition of success runs, let S_n, $M_n^{(k)}$ and L_n denote respectively the number of successes, the number of success runs with length k or more and the length of the longest success run in n Bernoulli trials, each with success probability p $(0 \le p \le 1)$. The probability of having a failure will be denoted with $q = 1 - p$ in the following.

Let us begin with a theorem that provides a first expression for the distribution of $M_n^{(k)}$:

Theorem 1 *If $M_n^{(k)}$ is the number of success runs with length k or more in n Bernoulli trials, we have*

$$P\left(M_n^{(k)} = x\right) = \sum_{m=x}^{\lfloor \frac{n+1}{k+1} \rfloor} (-1)^{m-x} \binom{m}{x} \sum_{y=m-1}^{n-mk} \binom{y+1}{m} \binom{n-mk}{y} p^{n-y} q^y \tag{1}$$

where k, n and x are positive integers.

Proof. Consider the following events

$$A_j = \{\text{A sequence of } k \text{ consecutive successes starts in } X_j\}$$

where X_j is the outcome of the jth trial and denote with $J_x = \{j_1, \ldots, j_x\}$ a subset of $\{1, \ldots, n\}$ containing exactly x different indices; we can write

$$P\left(M_n^{(k)} = x\right) = P\left(\bigcup_{j_1, j_2, \ldots, j_x} \left(A_{j_1} \cap A_{j_2} \cap \cdots \cap A_{j_x} \cap \bigcap_{j \notin J_x} \overline{A_j}\right)\right)$$

having denoted with $\overline{A_j}$ the complement of the set A_j. Thus by applying the inclusion-exclusion principle (see Feller, 1968, pag. 106) we obtain

$$P\left(M_n^{(k)} = x\right) = \sum_{m \ge x} (-1)^{m-x} \binom{m}{x} r_m \tag{2}$$

where r_m is given by

$$r_m = \sum_{j_1, \ldots, j_m} P\left(A_{j_1} \cap \cdots \cap A_{j_m}\right) = \sum_{y=m-1}^{n-mk} \sum_{j_1, \ldots, j_m} P\left(A_{j_1} \cap \cdots \cap A_{j_m}, S_n = n - y\right) \tag{3}$$

The bounds for the number of failures y can be easily obtained by noting that at least $m - 1$ failures are needed for separating the m success runs with length k or more starting in the positions $j_1, \ldots, j_m$. On the other hand, the realization of these runs requires at least mk successes.

Now, suppose without loss of generality that the indices $j_1, \ldots, j_m$ are ordered in an increasing way ($j_1 \le \cdots \le j_m$); according to the classical definition of run the sequences of n trials contained in the event A_j for $j > 1$ must have a failure as the $(j-1)$th outcome ($X_{j-1} = F$). It follows that the probability $P(A_{j_1} \cap \cdots \cap A_{j_m}, S_n = n - y)$ is nonnull only if

$$j_1 + k + 1 \le j_2, \quad \cdots \quad j_{m-1} + k + 1 \le j_m, \quad j_m + k - 1 \le n$$

Since $j_1 \ge 1$, by combining these inequalities we obtain that $r_m = 0$ for

$$1 + (k+1)(m-1) + k - 1 > n \quad \Longrightarrow \quad m > \frac{n+1}{k+1} \quad \Longrightarrow \quad m > \left\lfloor \frac{n+1}{k+1} \right\rfloor \tag{4}$$

having denoted with $\lfloor x \rfloor$ the integer not greater than x.

In the opposite case we note that

$$\sum_{j_1,\ldots,j_m} P\left(A_{j_1}\cap\cdots\cap A_{j_m}, S_n = n-y\right) = N_{m,y}p^{n-y}q^y \tag{5}$$

where $N_{m,y}$ is the number of different sequences of n Bernoulli trials having exactly $n-y$ successes and containing m success runs with length k or more. In fact, only these sequences, each of which has probability $p^{n-y}q^y$ of occurring, provide a nonnull contribution to the summation on the left hand side in (5).

A careful combinatorial reasoning leads to an explicit expression for $N_{m,y}$; in fact, if we consider the position of the y failures we have $\left(\begin{array}{c} y+1 \\ m \end{array}\right)$ different ways of placing m success runs of length k so that each of them is separated from the neighbors at least by a failure. Then we can put the remaining $n-y-mk$ successes into every configuration in $\left(\begin{array}{c} n-mk \\ y \end{array}\right)$ possible ways. Thus we obtain for $N_{m,y}$ the following expression

$$N_{m,y} = \left(\begin{array}{c} y+1 \\ m \end{array}\right)\left(\begin{array}{c} n-mk \\ y \end{array}\right) \tag{6}$$

By considering (6) and (5) the equation (3) for r_m becomes

$$r_m = \sum_{y=m-1}^{n-mk}\left(\begin{array}{c} y+1 \\ m \end{array}\right)\left(\begin{array}{c} n-mk \\ y \end{array}\right)p^{n-y}q^y$$

and (2) gives the desired expression (1) for $P(M_n^{(k)} = x)$ if we use the upper bound for m provided by (4). ∎

By interchanging the order of summation in (1) we have:

$$P\left(M_n^{(k)} = x\right) = \sum_{y=x-1}^{n-kx} p^{n-y}q^y \sum_{m=x}^{\min\left(y+1,\lfloor\frac{n-y}{k}\rfloor\right)} (-1)^{m-x}\left(\begin{array}{c} m \\ x \end{array}\right)\left(\begin{array}{c} y+1 \\ m \end{array}\right)\left(\begin{array}{c} n-mk \\ y \end{array}\right) \tag{7}$$

similar to the expression for $P(N_n^{(k)} = x)$ found in Godbole (1990) by employing the alternative definition of success run. The analogy between the two formulae is emphasized by setting $j = m-x$ in (7).

From theorem 1 we can directly obtain the relation for the distribution of the longest success run L_n in n Bernoulli trials. For this aim it is useful to enunciate the following

Lemma 1 *If k and n are positive integers, we have*

$$\sum_{m=0}^{\lfloor\frac{n-y}{k}\rfloor}(-1)^m\left(\begin{array}{c} y+1 \\ m \end{array}\right)\left(\begin{array}{c} n-mk \\ y \end{array}\right) = 0 \qquad \textit{for } 0\le y<\lfloor n/k\rfloor$$

Proof. Consider the function $f(x)$ given by

$$f(x) = \frac{\left(1-(1-x)^k\right)^{y+1}}{(1-x)^{n-y+1}} = \sum_{m=0}^{y+1}(-1)^m\left(\begin{array}{c} y+1 \\ m \end{array}\right)(1-x)^{mk-n+y-1}$$

and compute its yth derivative in the point $x = 0$

$$f^{(y)}(0) = \sum_{m=0}^{y+1} (-1)^m \binom{y+1}{m} \prod_{i=1}^{y} (n - mk - y + i) = y! \sum_{m=0}^{y+1} (-1)^m \binom{y+1}{m} \binom{n-mk}{y}$$

Now, the direct computation of the first y derivatives yields expressions containing a common multiplicative factor $(1-(1-x))^\nu$ where ν is a positive integer. Consequently we obtain

$$\sum_{m=0}^{y+1} (-1)^m \binom{y+1}{m} \binom{n-mk}{y} = \frac{1}{y!} f^{(y)}(0) = 0$$

Thus, consider the following two cases:

- when $0 \leq y < \lfloor (n+1)/(k+1) \rfloor$ we have $y + 1 < \lfloor (n-y)/k \rfloor$ and

$$\binom{y+1}{m} = 0 \qquad \text{for } y+1 < m \leq \lfloor (n-y)/k \rfloor \tag{8}$$

  consequently

$$\sum_{m=0}^{\lfloor \frac{n-y}{k} \rfloor} (-1)^m \binom{y+1}{m} \binom{n-mk}{y} = \sum_{m=0}^{y+1} (-1)^m \binom{y+1}{m} \binom{n-mk}{y} = 0 \tag{9}$$

- when $\lfloor (n+1)/(k+1) \rfloor \leq y < \lfloor n/k \rfloor$ we have $n - mk \geq 0$ for every $m \leq y+1$; then

$$\binom{n-mk}{y} = 0 \qquad \text{for } \lfloor (n-y)/k \rfloor < m \leq y+1$$

  and (9) is again verified. ∎

By taking into account (8) we can write

$$\sum_{m=0}^{\min\left(y+1, \lfloor \frac{n-y}{k} \rfloor\right)} (-1)^m \binom{y+1}{m} \binom{n-mk}{y} = \sum_{m=0}^{\lfloor \frac{n-y}{k} \rfloor} (-1)^m \binom{y+1}{m} \binom{n-mk}{y} \tag{10}$$

and by virtue of lemma 1 we obtain that the left hand side is null for $0 \leq y < \lfloor n/k \rfloor$. This result allows to find the correct expression for the distribution of L_n

Theorem 2 *If L_n denotes the length of the longest success run in n Bernoulli trials, we have*

$$P(L_n \leq k-1) = \sum_{y=\lfloor \frac{n}{k} \rfloor}^{n} p^{n-y} q^y \sum_{m=0}^{\lfloor \frac{n-y}{k} \rfloor} (-1)^m \binom{y+1}{m} \binom{n-mk}{y} \tag{11}$$

where k and n are positive integers.

Proof. Since $M_n^{(k)}$ denotes the number of success runs with length k or more, it follows from (1) that

$$\begin{aligned} P(L_n \geq k) &= \sum_{x=1}^{\lfloor \frac{n+1}{k+1} \rfloor} P\left(M_n^{(k)} = x\right) = \\ &= \sum_{x=1}^{\lfloor \frac{n+1}{k+1} \rfloor} \sum_{m=x}^{\lfloor \frac{n+1}{k+1} \rfloor} (-1)^{m-x} \binom{m}{x} \sum_{y=m-1}^{n-mk} \binom{y+1}{m} \binom{n-mk}{y} p^{n-y} q^y \end{aligned}$$

and by interchanging the summations on x and m:

$$\begin{aligned}
P\left(L_n \geq k\right) &= \sum_{m=1}^{\left\lfloor \frac{n+1}{k+1} \right\rfloor} \sum_{x=1}^{m} (-1)^{m-x} \binom{m}{x} \sum_{y=m-1}^{n-mk} \binom{y+1}{m} \binom{n-mk}{y} p^{n-y} q^{y} = \\
&= \sum_{m=1}^{\left\lfloor \frac{n+1}{k+1} \right\rfloor} (-1)^{m-1} \sum_{y=m-1}^{n-mk} \binom{y+1}{m} \binom{n-mk}{y} p^{n-y} q^{y}
\end{aligned}$$

being

$$\sum_{x=1}^{m} (-1)^{m-x} \binom{m}{x} = \sum_{x=0}^{m-1} (-1)^{x} \binom{m}{x} = \sum_{x=0}^{m} (-1)^{x} \binom{m}{x} - (-1)^{m} = (-1)^{m-1}$$

Now, we note that

$$P\left(L_n \leq k-1\right) = 1 - P\left(L_n \geq k\right) = \sum_{m=0}^{\left\lfloor \frac{n+1}{k+1} \right\rfloor} (-1)^{m} \sum_{y=m-1}^{n-mk} \binom{y+1}{m} \binom{n-mk}{y} p^{n-y} q^{y}$$

and by interchanging the order of summation

$$P\left(L_n \leq k-1\right) = \sum_{y=0}^{n} p^{n-y} q^{y} \sum_{m=0}^{\min\left(y+1, \left\lfloor \frac{n-y}{k} \right\rfloor\right)} (-1)^{m} \binom{y+1}{m} \binom{n-mk}{y} \tag{12}$$

In fact the inequality $y \geq m-1$ gives the upper bound $y+1$ for m while $y \leq n - mk$ leads to $m \leq \lfloor (n-y)/k \rfloor$. But, by virtue of (10) and lemma 1 we obtain from (12) the desired relation (11). ∎

Theorem 2 provides the well known expression for $P(L_n \leq k-1)$ already obtained by Burr and Cane (1961) and Godbole (1990) with other methods. From this result also the formulae for $P(L_n \leq k, S_n = r)$ and $P(L_n \leq k, S_n = r)$ presented in Gibbons (1971) follows directly.

Incidentally, equation (12) could be obtained by setting $x = 0$ in (7); in this way the achievement of (11) would have been shorter. Unfortunately, the proof of theorem 1 only holds for positive values of x and thus the passage above would not be theoretically acceptable.

3 Simplified expressions for some success run distributions

From equations (1) and (11) obtained for the distributions of $M_n^{(k)}$ and L_n respectively follow some interesting simplified expressions. They contain only a single summation of ordinary (first order) binomial coefficients and therefore their corresponding computation time is considerably lowered.

Theorem 3 *If $M_n^{(k)}$ is the number of success runs with length k or more in n Bernoulli trials, we have*

$$P\left(M_n^{(k)} = x\right) = \sum_{m=x}^{\left\lfloor \frac{n+1}{k+1} \right\rfloor} (-1)^{m-x} \binom{m}{x} p^{mk} q^{m-1} \left(\binom{n-mk}{m-1} + q \binom{n-mk}{m} \right) \tag{13}$$

where k, n and x are positive integers.

Proof. By setting $j = n - mk - y$ in (1), we obtain

$$P\left(M_n^{(k)} = x\right) =$$
$$= \sum_{m=x}^{\lfloor \frac{n+1}{k+1} \rfloor} (-1)^{m-x} \binom{m}{x} \sum_{j=0}^{n-mk-m+1} \binom{n-mk-j+1}{m} \binom{n-mk}{j} p^{mk+j} q^{n-mk-j} =$$
$$= \sum_{m=x}^{\lfloor \frac{n+1}{k+1} \rfloor} (-1)^{m-x} \binom{m}{x} p^{mk} q^{n-mk} \sum_{j=0}^{n-mk-m+1} \binom{n-mk-j+1}{m} \binom{n-mk}{j} (p/q)^j \quad (14)$$

Now, if we make use of the Pascal triangle identity, we have

$$\sum_{j=0}^{n-mk-m+1} \binom{n-mk-j+1}{m} \binom{n-mk}{j} (p/q)^j =$$
$$= \sum_{j=0}^{n-mk-m+1} \left(\binom{n-mk-j}{m-1} + \binom{n-mk-j}{m} \right) \binom{n-mk}{j} (p/q)^j =$$
$$= \sum_{j=0}^{n-mk-m+1} \left(\binom{n-mk-j}{n-mk-m+1-j} + \binom{n-mk-j}{n-mk-m-j} \right) \binom{n-mk}{j} (p/q)^j =$$
$$= \binom{n-mk}{m-1} (1/q)^{n-mk-m+1} + \binom{n-mk}{m} (1/q)^{n-mk-m} \quad (15)$$

In the last passage the following relation has been employed (see Feller, 1968, pag. 63):

$$\sum_{\nu \geq 0} \binom{h}{\nu} \binom{h-\nu}{r-\nu} t^{\nu} = \binom{h}{r} (1+t)^r$$

which holds for r, h non-negative integers and for every real number t.

By substituting (15) in (14) we obtain the desired relation (13). ∎

From (13) it is possible to obtain the corresponding simplified expression for the distribution of the longest success run L_n. This formula has already been found by Lambiris and Papastavridis (1985) and Hwang (1986) in the study of reliability for consecutive-k-out-of-n systems.

Corollary 1 *If L_n denotes the length of the longest success run in n Bernoulli trials, we have*

$$P(L_n \leq k-1) = \sum_{m=0}^{\lfloor \frac{n+1}{k+1} \rfloor} (-1)^m p^{mk} q^{m-1} \left(\binom{n-mk}{m-1} + q \binom{n-mk}{m} \right) \quad (16)$$

where k and n are positive integers.

Proof. It is sufficient to proceed as in the first part of the proof of theorem 2 by noting that

$$P(L_n \leq k-1) = 1 - P(L_n \geq k) = 1 - \sum_{x=1}^{\lfloor \frac{n+1}{k+1} \rfloor} P\left(M_n^{(k)} = x\right) =$$
$$= 1 - \sum_{x=1}^{\lfloor \frac{n+1}{k+1} \rfloor} \sum_{m=x}^{\lfloor \frac{n+1}{k+1} \rfloor} (-1)^{m-x} \binom{m}{x} p^{mk} q^{m-1} \left(\binom{n-mk}{m-1} + q \binom{n-mk}{m} \right) \quad ∎$$

The simplified formulae (13) and (16) per $P\left(M_n^{(k)}=x\right)$ and $P\left(L_n \leq k-1\right)$ can be used for obtaining expressions with single summation of other interesting probability distributions. As an example let us consider the kth order negative binomial random variable $\mathrm{NB}_{k,r}$ defined as the waiting time till the rth success run of length k or more (introduced by Philippou, Georghiou and Philippou, 1983, with the alternative definition of run).

In case of classical definition of success run we have the following

Theorem 4 *The random variable* $\mathrm{NB}_{k,r}$ *is characterized by the following probability distribution*

$$P\left(\mathrm{NB}_{k,r}=x\right)=$$
$$=\sum_{m=r}^{\left\lfloor\frac{x+1}{k+1}\right\rfloor}(-1)^{m-r}\binom{m-1}{r-1}p^{mk}q^{m-1}\left(\binom{x-mk-1}{m-2}+q\binom{x-mk-1}{m-1}\right) \tag{17}$$

Proof. By definition of $\mathrm{NB}_{k,r}$ every sequence of n Bernoulli trials belonging to the event $\{\mathrm{NB}_{k,r}=x\}$ must end with k successes preceded by a failure. Thus, we have

$$P\left(\mathrm{NB}_{k,r}=x\right)=p^k q\cdot P\left(M_{x-k-1}^{(k)}=r-1\right)$$

and by using the expression (13) for $P\left(M_{x-k-1}^{(k)}=r-1\right)$ we obtain

$$P\left(\mathrm{NB}_{k,r}=x\right)=p^k q\sum_{m=r-1}^{\left\lfloor\frac{x-k}{k+1}\right\rfloor}(-1)^{m-r+1}\binom{m}{r-1}p^{mk}q^{m-1}\cdot$$
$$\cdot\left(\binom{x-(m+1)k-1}{m-1}+q\binom{x-(m+1)k-1}{m}\right)=$$
$$=\sum_{m=r}^{\left\lfloor\frac{x+1}{k+1}\right\rfloor}(-1)^{m-r}\binom{m-1}{r-1}p^{mk}q^{m-1}\left(\binom{x-mk-1}{m-2}+q\binom{x-mk-1}{m-1}\right) \qquad \blacksquare$$

This theorem also allows to obtain the formula for the probability distribution of the kth order geometric random variable G_k; it is sufficient to set $r=1$ in (17)

$$P\left(G_k=x\right)=\sum_{m=1}^{\left\lfloor\frac{x+1}{k+1}\right\rfloor}(-1)^{m-1}p^{mk}q^{m-1}\left(\binom{x-mk-1}{m-2}+q\binom{x-mk-1}{m-1}\right)$$

In this case the two expressions deriving from different definitions of success run coincide (Godbole, 1990).

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